GHRaisoni International Skill Tech University, Pune

SYSTEM & UML DIAGRAMS

AI Chatbot for College · BBA Computer Applications

1. Use Case Diagram

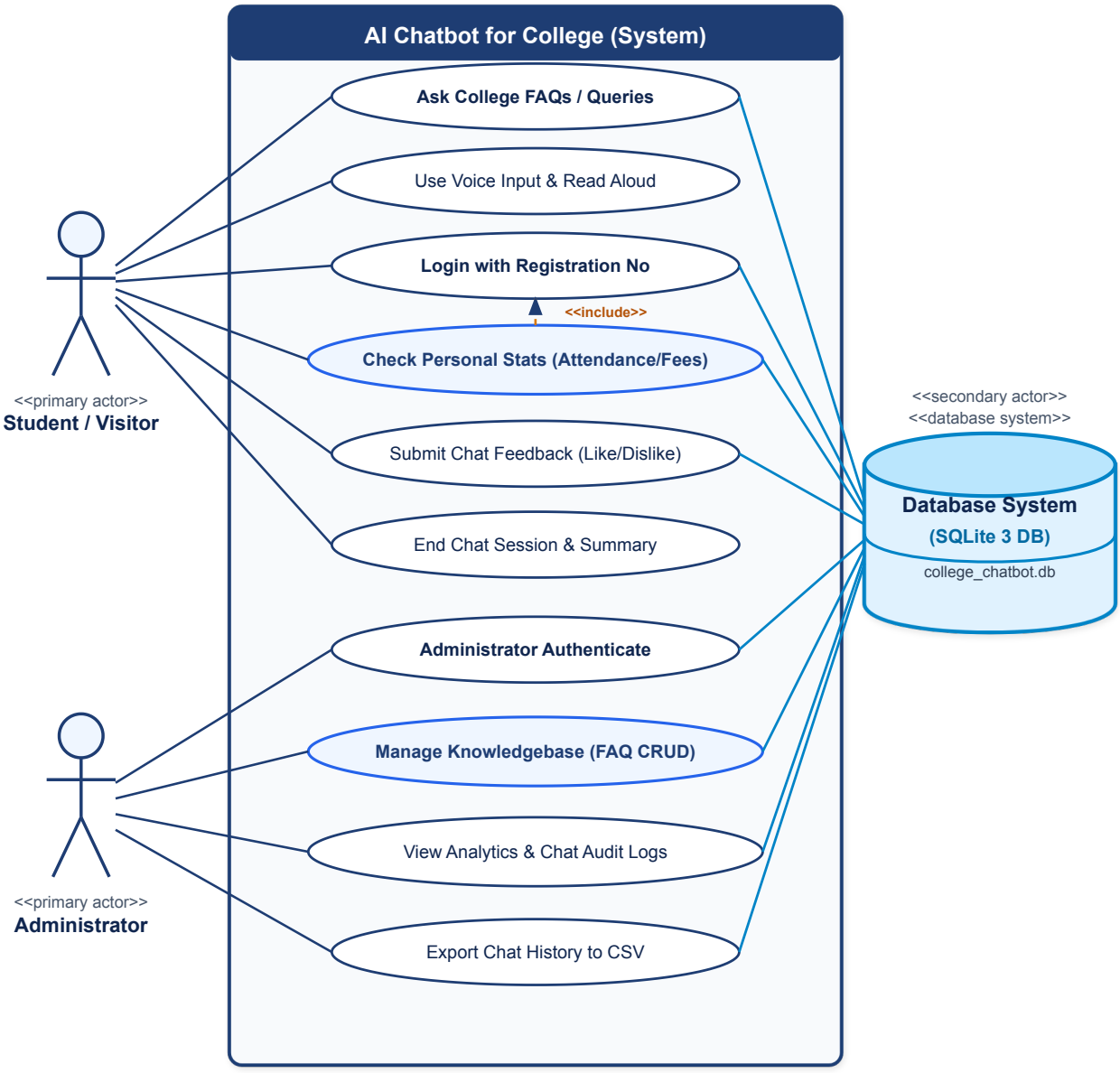


Fig 1: Use Case Diagram — primary actors (Student & Administrator), system boundary, and broad secondary database cylinder (SQLite 3 DB) with data-tier associations.

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2. Class Diagram (With Full Relations & Multiplicities)

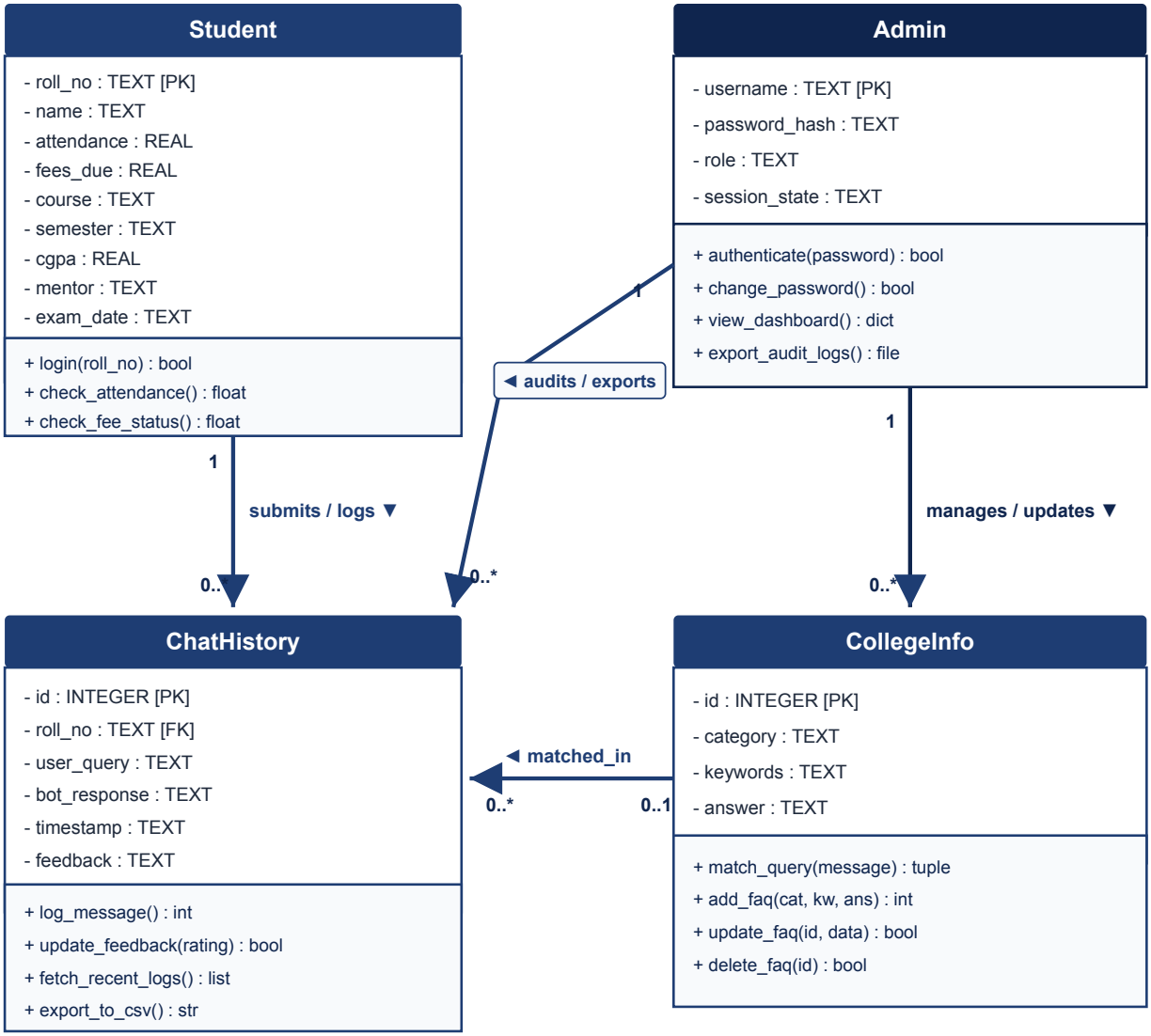


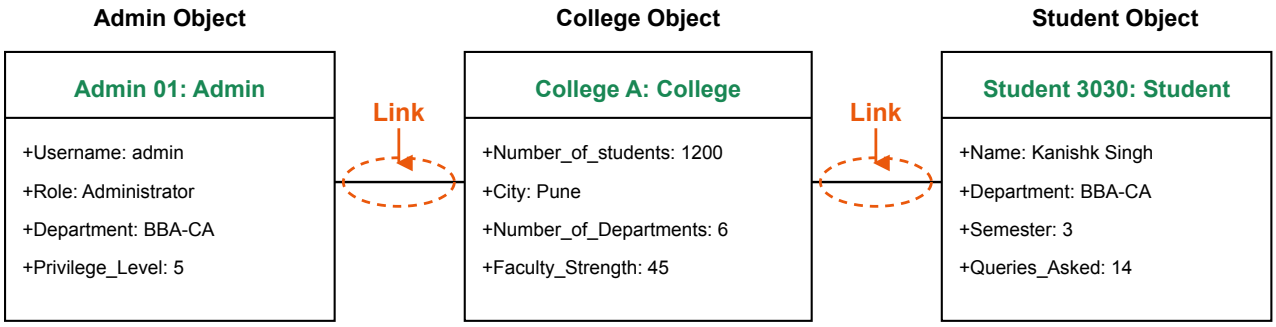
Fig 2: Class Diagram — core entities, attributes, operations, and 100% solid association arrows with multiplicities and role verbs.

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3. Object Diagram (Runtime Instances & Slot Values)

An object diagram using a link and 3 objects



An object of class Admin and an object of class Student are linked to an object of class College.

Object Diagrams | Unified Modeling Language (UML) ⚙ GfG

Fig 3: Object Diagram — runtime object instances (Admin Object, College Object, Student Object) connected via formal UML links matching GeeksforGeeks standard notation.

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4. Sequence Diagram — Query, Feedback & Session Termination

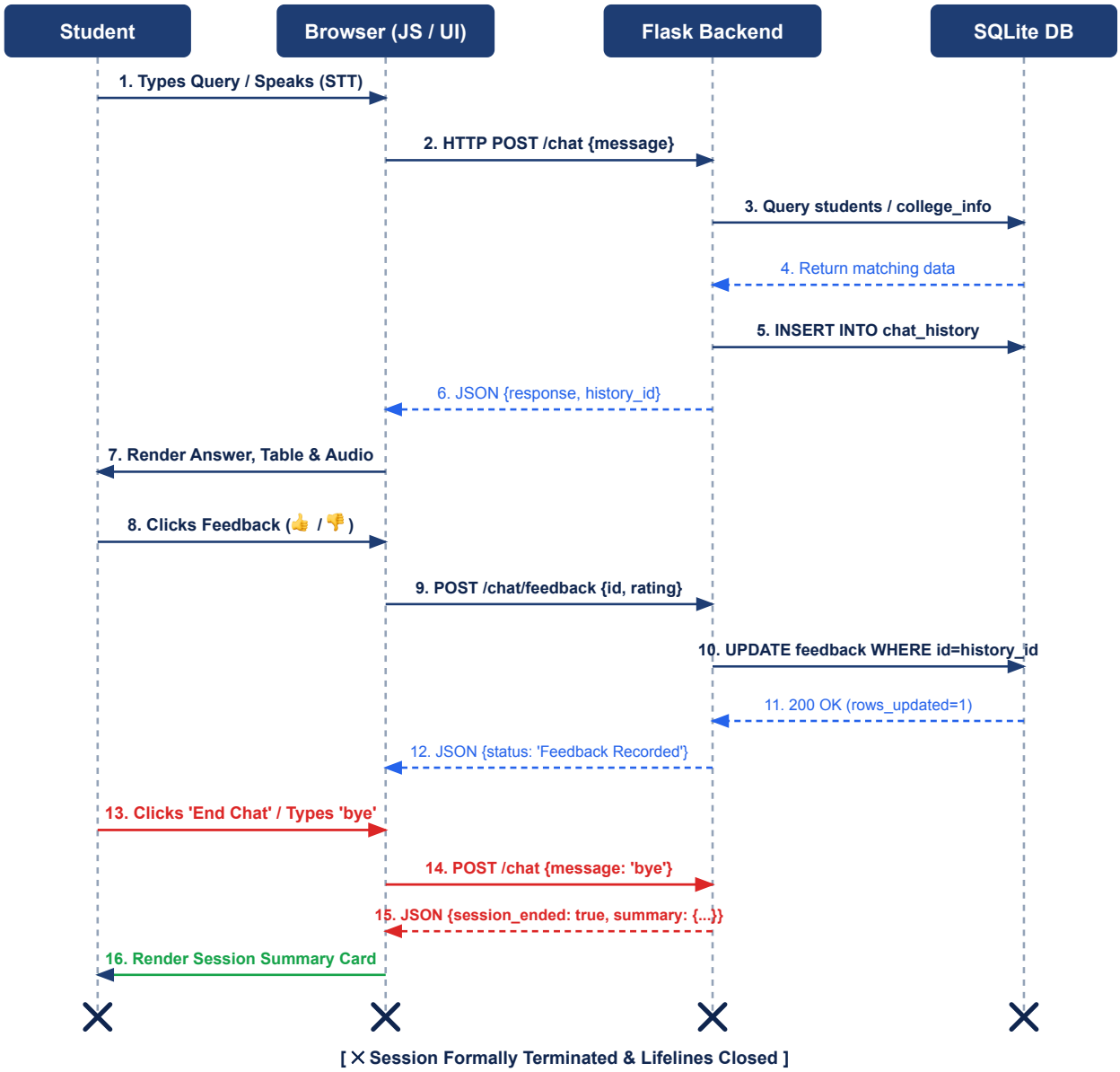


Fig 4: Sequence Diagram — complete query, feedback ACK, explicit End Chat flow, and standard UML lifeline termination (X).

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5. Activity Diagram — Chatbot Response & Session Flow

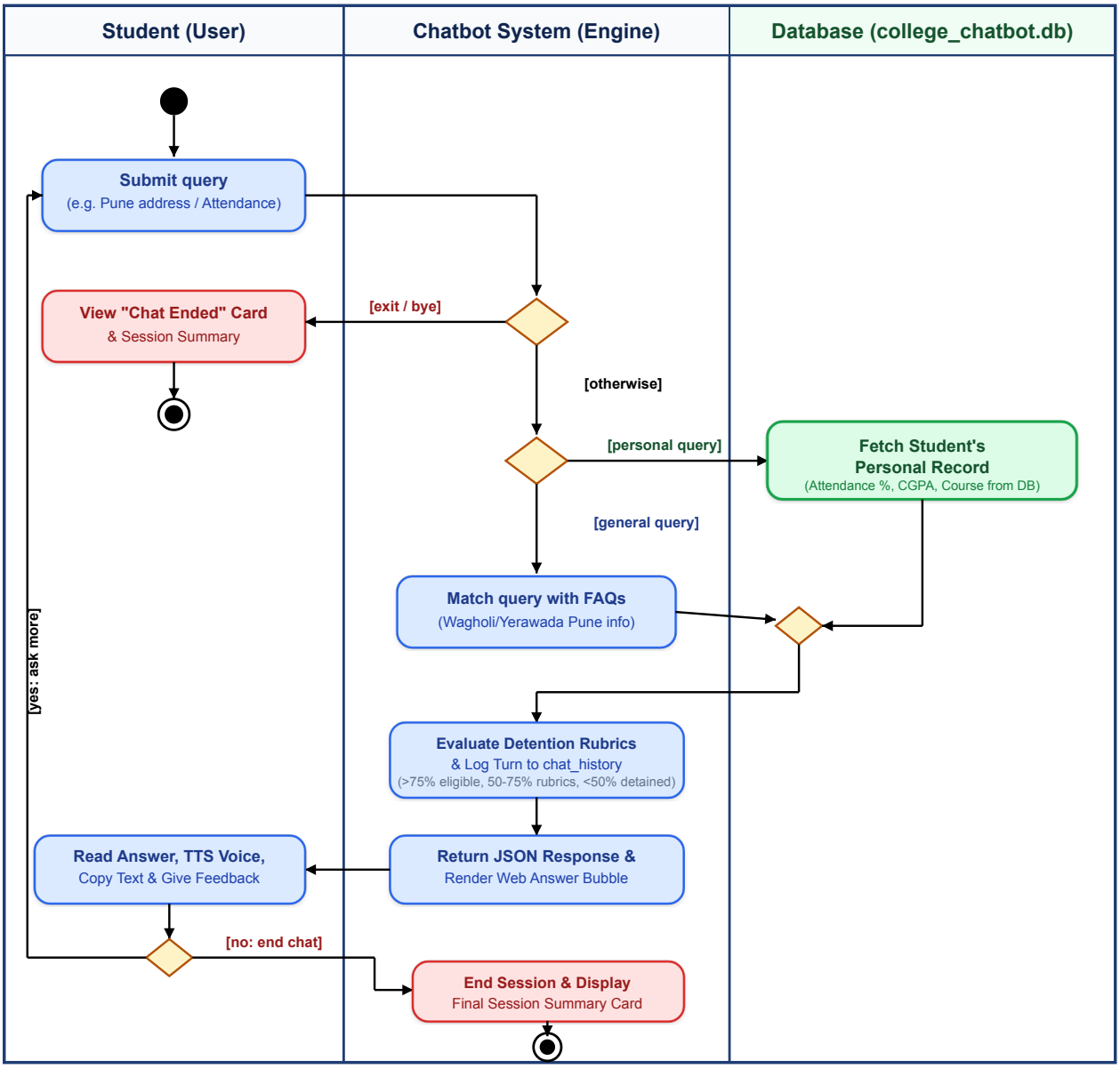
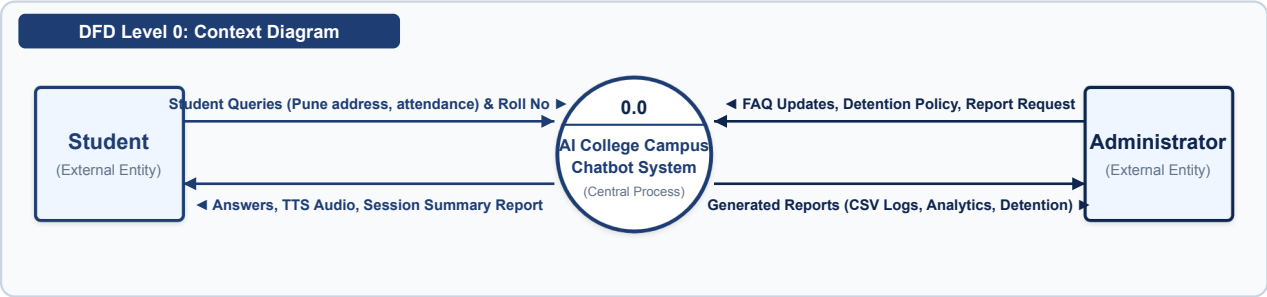


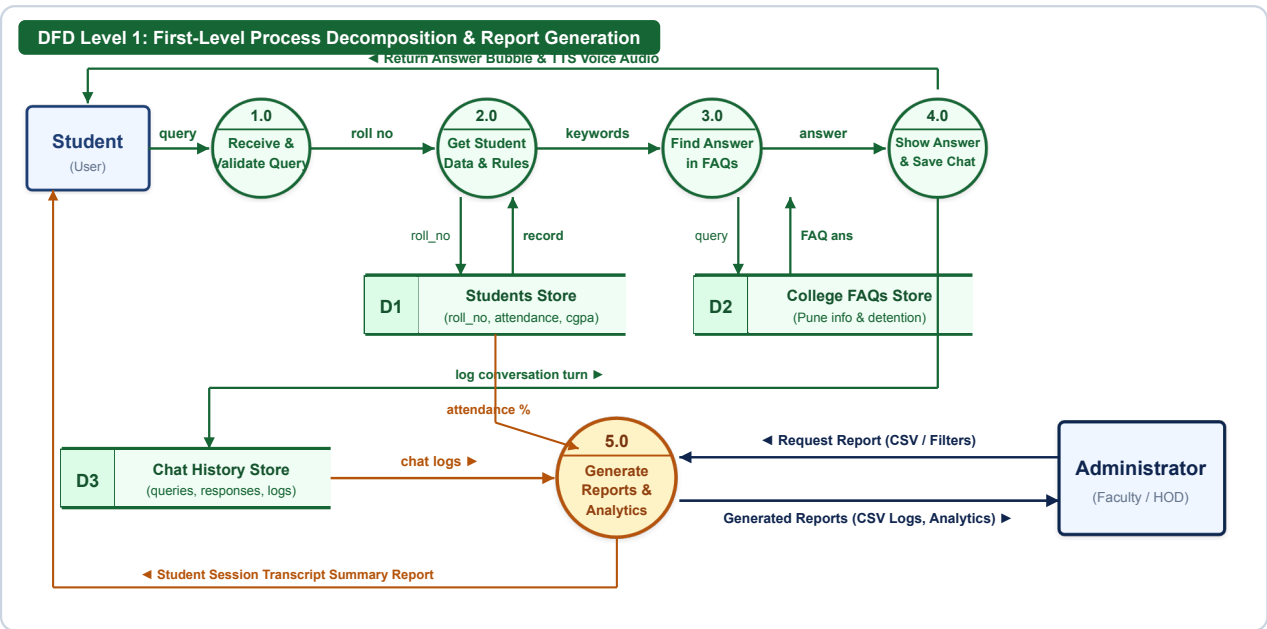
Fig 5: Activity Diagram — 3-Swimlane architecture (Student, Chatbot System, Database) explicitly showing database fetching of student personal records, FAQ matching, and controlled session loop with explicit End states.

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6. Data Flow Diagram (DFD Level 0 Context & Level 1 Process Breakdown with Report Generation)





Standard DFD Notation & Symbol Conventions (Gane-Sarson & DeMarco)

Symbol Shape	DFD Element	Standard Academic Definition	Implementation in Our Project
▭	**External Entity**	Origin/destination of data outside boundary (Square)	**Student (User) & Administrator (Faculty/HOD)**
○	**Process (Bubble)**	Transforms input data into output data (Circle with ID) Calculates, retrieves, formats, or generates reports	**1.0 to 4.0 Query Handlers & 5.0 Generate Reports** Query answering & administrative CSV/analytics reporting
D ▭	**Data Store** *(Open-Ended)*	Repository of data at rest (Open parallel lines) **MANDATORY RULE: Right side must be open; NEVER closed**	**D1 Students, D2 College FAQs, D3 Chat History** Persistent SQLite tables feeding search & report generation
→	**Data Flow**	Directed solid arrow showing data path & payload Transfers packets between entities, processes, and stores	**query, record, FAQ answer, generated reports CSV** Directional vectors with explicit noun-phrase labels

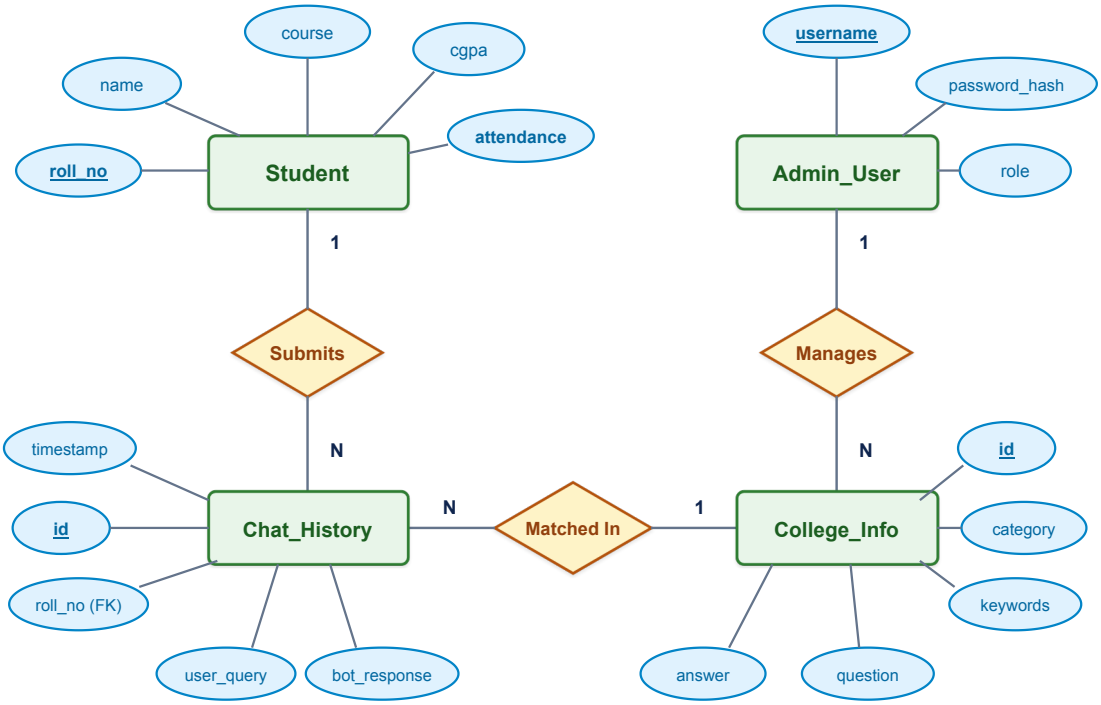
Fig 6: Data Flow Diagram (DFD) — Properly formatted Level 0 Context Diagram (System boundary with Student & Administrator) plus Level 1 Process Decomposition (1.0 to 4.0 Query Handlers & 5.0 Generate Reports & Analytics) with standard open-ended data stores (D1 Students, D2 FAQs, D3 Chat History) and formal academic DFD symbols reference table.

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7. Entity-Relationship Diagram (ERD — Database Architecture)

ER Diagram (Classic Notation): AI Chatbot for College



Schema & Project Knowledge Base Defense:

- **Student Entity:** Stores attendance %, CGPA & evaluates 4-tier detention policy.
- **College_Info Entity:** Contains Pune campuses (Wagholi & Yerawada) & exam rules.
- **Chat_History Entity:** Logs each turn with student roll_no FK, query & bot answer.
- **Admin_User Entity:** RBAC authentication for updating college FAQs & detention data.

Verified in 3NF Normalization: Zero transitive dependencies & lossless join preservation.

Key / Legend:

- Entity (Table)
- Attribute (Column) — id = Primary Key
- Relationship
- Link / Cardinality (1:N)

Fig 7: Entity-Relationship Diagram (ERD — Classic Notation) — 4 core entities (Student, Admin_User, Chat_History, College_Info) with attributes, cardinalities (1:N, N:1), and official symbol key.

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8. Deployment Diagram (Hosting Topology & Artifact Architecture)

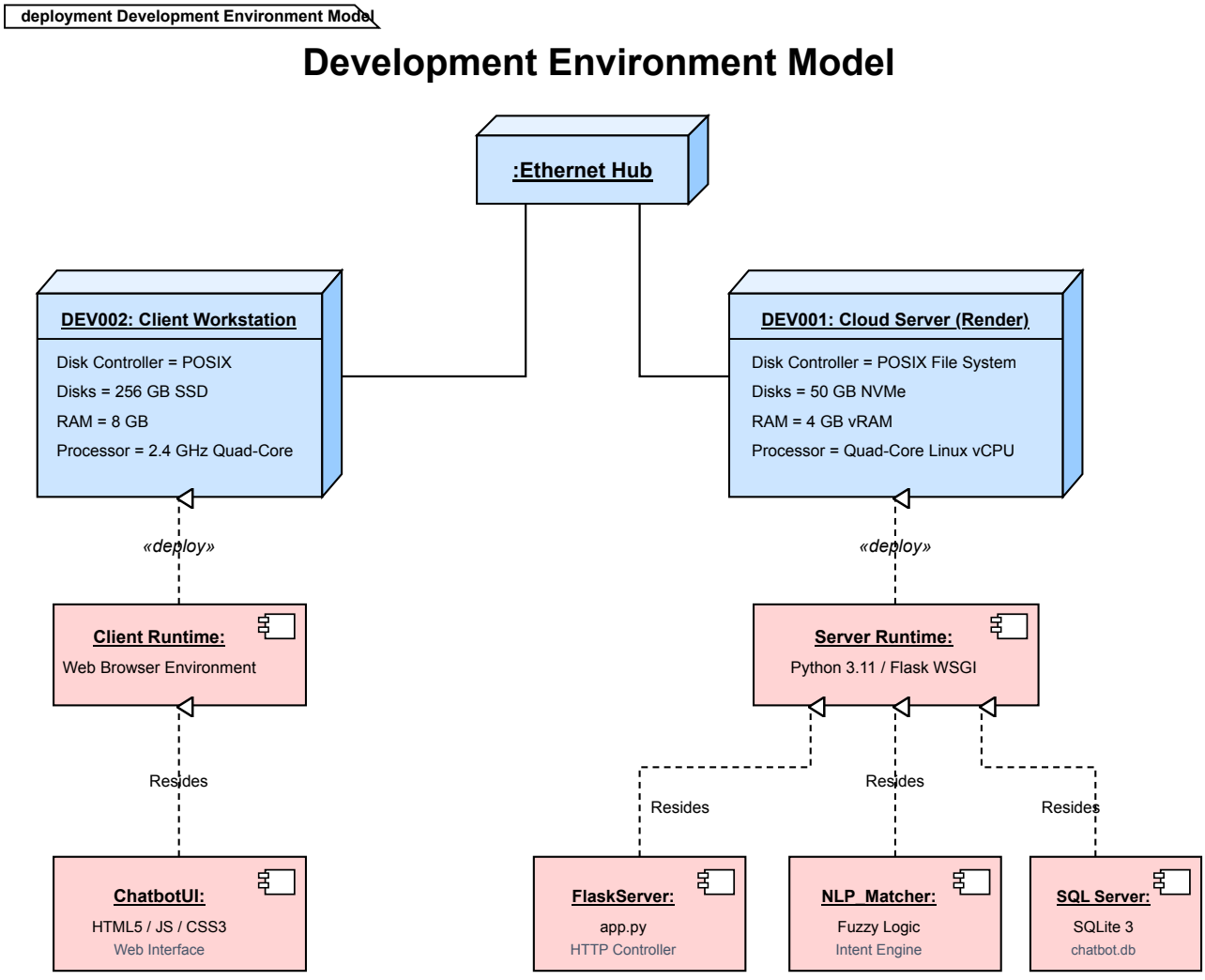


Fig 8: Deployment Diagram — Development Environment Model with :Ethernet Hub, 3D hardware nodes with specifications, «deploy» dependency arrows, and resident software component boxes.